PANELSYNC

MVP Build Document — v2.0

This document defines exactly what gets built in the PanelSync MVP, in what order, and what is deliberately deferred. It is a build document, not a vision document. Every decision here is made in service of proving the core hypothesis as fast as possible with a lean, solo-built bootstrap.

The Hypothesis	*Is PanelSync a better place to write a comic and organize a fictional world than Google Docs, Apple Notes, and a folder of reference images?*

1. MVP Goal & Success Definition

The MVP succeeds if a writer and their collaborator can move their entire comic workflow — scripts, characters, locations, and timelines — into PanelSync and feel more organized, more aligned, and less frustrated than they were before.

Success is not feature completeness. Success is one workflow, end to end, that feels genuinely better than what the user had before.

1.1 Success Metrics

- At least 10 real users complete a full working session — writing script pages, adding characters or locations, and creating at least one timeline — without prompting
- At least 7 of those 10 say they would use it over their current tools
- At least one collaborating pair uses it together for a full week and reports feeling more organized
- Zero critical bugs that block core writing or organizing workflows
- Web app loads and is usable in under 3 seconds
- iPad Apple Pencil latency is imperceptible during normal sketching

2. What's In the MVP

Scope Rule	*If a feature does not directly help a writer write scripts, organize their world, or collaborate with a partner — it is not in the MVP.*

2.1 Feature Status Table

Feature	MVP?	Notes
Worlds Dashboard — create and manage Worlds	**YES**	Entry point for the whole product.
World hierarchy — World, Series, Issues, Pages	**YES**	Core structure. Everything lives here.
Script editor — comic-native block types	**YES**	Panel, Scene Heading, Description, Dialogue, Caption, SFX.
Script editor — distraction-free mode	**YES**	Full screen writing, everything hidden.
Script editor — outline view	**YES**	Collapsed to panel headings only.
Script editor — auto-save	**YES**	Debounced 1.5s. Subtle save indicator.
Storyboard — blank drawable canvas per script page	**YES**	One canvas per page, always paired. No auto-generation in v1.
Storyboard — freehand drawing on iPad	**YES**	Pencil, eraser, pressure sensitivity, undo/redo.
Storyboard — reference image import + trace	**YES**	Import image, set opacity, sketch over it.
Storyboard — script breakdown grid	**YES**	All pages as thumbnails with status indicators.
Drawing on character pages (iPad)	**YES**	Same Skia canvas, same tools. Character design workflow.
Drawing on location pages (iPad)	**YES**	Same Skia canvas, same tools. Location design workflow.
World Bible — Characters	**YES**	Profile, rich text notes, reference images, sketch layer.
World Bible — Locations	**YES**	Profile, rich text notes, reference images, sketch layer.
Timeline — create multiple named timelines	**YES**	Story, reference, character arc, production intent tags.

Timeline — events with point or range tags	YES	Title, date label, description, character tags.
Timeline — named ranges	YES	Group events into jumpable named spans.
Timeline — zoom, density toggle, color coding	YES	Full navigation toolkit.
Timeline — jump to character, date, range	YES	Fast navigation across long timelines.
Timeline — pin timelines side by side	YES	Compare any two timelines simultaneously.
Notes — folders, rich text, quick-capture, search	YES	World-level notes. Attached notes in v2.
Async collaboration — invite collaborators	YES	Editor and Viewer roles. Last-write-wins sync.
Page status tracking — draft, review, locked, complete	YES	Per-page status visible in breakdown grid.
Comments on script lines	YES	Threaded text comments.
Read-only share link	YES	No account required to view.
Script PDF export	YES	Industry-standard format.
World Bible — Factions, Objects, Lore	NO	v2 — Characters and Locations cover MVP needs.
Timeline lenses (reader knowledge, character knowledge)	NO	v2 — data model ready, lenses come later.
Storyboard auto-generation from script	NO	v2 — data model and page structure designed for it.
The Chalkboard	NO	v2 — powerful but not needed to prove the hypothesis.
Relationship Map	NO	v2.
Mood / Tone Board	NO	v2.
Pacing View	NO	v2.
Continuity Tracker	NO	v2 — tags built in MVP, detection comes later.
Real-time collaborative sync (Yjs)	NO	v2 — async is enough for MVP.
Series Bible PDF export	NO	v2.
Storyboard PDF export	PARTIAL	Basic server-side render. No layer control in v1.
Reading mode, Presentation mode, Print layout	NO	v2.
The Letterer	NO	v2.
Version history / named snapshots	NO	v2 — auto-save only in v1.
Offline mode	NO	v2.
Android tablet	NO	v2 — same Expo codebase, primarily testing work.

3. Platform & Stack

3.1 The Unified Approach

PanelSync is one product that runs on two surfaces from a single codebase. The iPad is the full product. The web is the same product optimized for a keyboard and mouse, with drawing canvas depth reduced. They look and behave as similarly as possible — platform differences are limited to input method and canvas capability.

Design Principle	*Anything you can do in the script editor on web, you can do on iPad. Platform differences are drawing depth only — not feature access.*

3.2 Technology Stack

Layer	Technology	Rationale
App framework	Expo SDK (React Native + Expo for Web)	Single codebase runs natively on iPad and in browser. Future Android with no rewrite.
Script editor	Custom rich text engine with portable ProseMirror adapter	Same editor component on both platforms. Custom block types for all comic block types.
Drawing canvas	React Native Skia (iPad native only)	High-performance canvas with Apple Pencil pressure sensitivity. Not loaded on web in v1.
Shared UI system	NativeWind (Tailwind for React Native)	Utility-class styling across native and web from one set of class names.
Navigation	Expo Router	File-based routing, works identically on native and web.
Backend API	Node.js + Fastify	Lightweight REST API shared by both surfaces.
Database	PostgreSQL	All relational data — projects, users, pages, notes, timeline events.
File storage	Cloudflare R2 (S3-compatible)	Sketch data, reference images, exported PDFs. No egress fees.
Auth	Expo Auth + custom JWT	Email/password. OAuth (Google) as stretch goal.

Sync	REST polling on open	Loose sync — no WebSocket in MVP. Page fetches latest on open.
Hosting — web	Vercel	Zero-config Expo web export deployment. Free tier for MVP.
Hosting — backend	Railway or Render	Postgres + Node. Free/low-cost tier.
PDF export	Puppeteer (server-side)	Headless Chrome renders script or storyboard to PDF.

4. Core Data Model

The MVP data model is intentionally minimal. Every entity below is load-bearing for the core workflow. Nothing is added speculatively. The model is designed to accommodate v2 features — timeline lenses, continuity tracking, storyboard auto-generation — without a rewrite.

4.1 Entities

- User — id, email, password hash, display name, avatar
- World — id, owner_id, name, cover_image, created_at
- WorldMember — world_id, user_id, role (owner / editor / commenter / viewer)
- Series — id, world_id, name, arc_notes, cover_image
- Issue — id, series_id, number, title, notes
- Page — id, issue_id, page_number, script_content (JSON), status (draft/review/locked/complete), updated_at, updated_by
- StoryboardPage — id, page_id, sketch_data (JSON), reference_image_url, reference_opacity
- Panel — id, page_id, panel_number, size_tag, keywords[] — prepared for v2 auto-generation
- Character — id, world_id, name, role, appearance, backstory, color, knowledge_gaps (JSON)
- CharacterSeriesOverlay — character_id, series_id, notes
- Location — id, world_id, name, description, time_period, color
- LocationSeriesOverlay — location_id, series_id, notes
- Timeline — id, world_id, name, intent (story/reference/character_arc/production)
- TimelineEvent — id, timeline_id, title, date_label, description, character_ids[], location_id, tag_type (point/range), range_start_label, range_end_label, color, sort_order
- TimelineRange — id, timeline_id, name, start_event_id, end_event_id
- Note — id, world_id, series_id (nullable), issue_id (nullable), title, content (JSON), is_private, updated_at

- NoteFolder — id, world_id, name, parent_folder_id
- Comment — id, target_type, target_id, user_id, content, resolved, created_at
- AssetTimelineTag — id, asset_type, asset_id, timeline_event_id, tag_type (point/range)

4.2 Key Design Decisions

- StoryboardPage is separate from Page — sketch data never overwrites script content
- Panel entity exists in MVP even without auto-generation — the panel_number and size_tag fields are written by the script parser and read by the v2 storyboard generator
- knowledge_gaps on Character is a JSON field that stores narrative knowledge gap tags — ready for the v2 Continuity Tracker to read
- AssetTimelineTag is a polymorphic join table — any asset type (character, location, sketch, note, page) can be tagged to any timeline event without schema changes
- CharacterSeriesOverlay and LocationSeriesOverlay allow series-specific notes to layer on world-level entries without overwriting them

5. Feature Specifications

5.1 Script Editor (iPad + Web — identical)

The script editor is a single shared component that runs on both iPad and web. Every block type, keyboard shortcut, and formatting behavior works identically on both platforms.

- Custom block types: PanelBlock, SceneHeadingBlock, DescriptionBlock, DialogueBlock, CaptionBlock, SFXBlock
- PanelBlock auto-numbers on creation; renumbers all subsequent panels on deletion
- DialogueBlock character name field has autocomplete from the series character registry
- Inline size tags parsed from DescriptionBlock text — typed inline, no special UI
- Cmd/Ctrl + Enter — insert new panel block
- Tab — cycle through block types within a panel
- Split view — script left (60%), storyboard preview right (40%). On iPad, the right panel is the live Skia drawing canvas. On web, it is a read-only storyboard preview in v1.
- Distraction-free mode — Cmd/Ctrl + Shift + F. Full screen, sidebar and preview hidden.
- Outline view — all blocks collapsed to panel headings only. Toggle from the view menu.
- Auto-save — debounced 1.5s. Subtle animated save indicator in the toolbar.
- On iPad — software keyboard auto-dismisses when the user taps the storyboard canvas side, transitioning naturally from writing to drawing mode

5.2 Storyboard Canvas (iPad native)

- One blank Skia canvas per script page, always paired
- Panel grid guides as a toggleable overlay — helps the artist divide the page before sketching
- Script for the current page accessible as a collapsible reference panel on the left
- Pencil tool with Apple Pencil pressure sensitivity — pressure maps to stroke width
- Eraser, stroke weight slider, stroke color (black, dark grey, red for annotation)
- Two-finger tap: undo. Three-finger tap: redo.
- Pinch to zoom the full page. Double-tap to enter focused mode on a panel area.
- Toolbar auto-hides after 3 seconds of drawing; tap to reveal
- Page strip at bottom — thumbnail navigation between pages in the current issue
- Reference image import — import from photo library or files, set opacity, sketch over
- Auto-save sketch data every 30 seconds and on navigation away from the page

5.3 World Bible — Characters & Locations (iPad + Web)

- Character profile fields: name, role, appearance description, backstory, series appearances, color (for proxy markers and timeline color coding)
- Location profile fields: name, description, time period, series appearances, color
- Rich text notes section on every entry
- Reference image import — attach images to any entry
- Sketch layer on iPad — draw directly on the character or location page with Apple Pencil. Reference images traceable.
- Series overlays — per-series notes that layer on top of the world-level entry
- Knowledge gap tagging on character entries — 'this character does not know X at this point in the story' — stored as JSON, read by v2 Continuity Tracker
- Browse by type (all Characters, all Locations), filter by series, global search

5.4 Timeline (iPad + Web — identical)

- Create as many timelines as needed within a World
- Intent tag on creation — story, reference, character arc, or production
- Events: title, date label (free text), description, character tags, location tag, point or range marker
- Named ranges — group any span of events with a name, jumpable from the range index
- Chronological order by default; drag-to-reorder available
- Zoom — scale from centuries to hours
- Density toggle — compact (titles only) or expanded (full details)
- Color coding — by character, location, or event type
- Jump to character, date, named range
- Pin any two timelines side by side for comparison
- Asset tagging — any character, location, note, or sketch can be tagged to a timeline event or range

5.5 Notes (iPad + Web — identical)

- World-level notes organized in folders with drag-and-drop
- Rich text editor — bold, italic, headings, bullet lists, inline code
- Quick-capture — floating button on iPad, Cmd/Ctrl + Shift + N on web
- Full-text search across all notes in a World
- Private notes — visible only to the creator

5.6 Collaboration (iPad + Web)

- Invite collaborators by email — they receive a link to create an account and join the World
- Roles: Owner, Editor, Viewer
- Last-write-wins sync with a visible 'last edited by [name] [time]' indicator on every page
- Update banner when a page has been edited since the user last opened it
- Page status tracking — draft, in review, locked, complete — set per page from the script toolbar
- Script breakdown grid — all pages as thumbnails with status color codes
- Threaded text comments on any script block — leave, reply to, and resolve comments
- Read-only share link — anyone can view the storyboard without an account

6. Build Order & Sprint Plan

Estimated at 2-week sprints for a single technical founder working part-time (roughly 20 hours/week). The unified Expo codebase front-loads setup work in Sprints 1-2 but eliminates platform duplication from Sprint 3 onward.

Sprint	Deliverable	Success Criteria
1–2	Foundation: Expo monorepo, NativeWind design system, Expo Router navigation, shared component architecture, backend API skeleton, database schema, auth (email + password), Worlds Dashboard	App runs on iPad and browser with the same shell. Can sign up, sign in, create a World, and see the Worlds Dashboard on both platforms.
3–4	Hierarchy: Series, Issue, Page CRUD. Sidebar navigation. Contextual sidebar that shifts as you navigate deeper. Script page and storyboard page always created as a pair.	Can create a Series, add Issues, add Pages, navigate the sidebar on both platforms. Each page has a paired storyboard canvas.

5–6	Script editor: shared rich text engine, all six comic block types, auto-numbering, Tab cycling, auto-save, distraction-free mode, outline view. Same component on iPad and web.	Can write a full multi-panel script page on both iPad and web with all block types working correctly.
7–8	Character registry: character profiles, series overlays, autocomplete in DialogueBlock. Location profiles. World Bible browse and search. Knowledge gap tag field on character entries.	Can create characters and locations, write dialogue with character autocomplete, browse the World Bible on both platforms.
9–10	iPad drawing: Skia canvas on storyboard pages, character pages, and location pages. Apple Pencil pressure, eraser, toolbar, undo/redo, reference image import, opacity control, auto-save sketch data.	Can sketch on any drawable surface on iPad with Apple Pencil. Sketches persist across sessions.
11–12	Timeline: create timelines with intent tags, add events (point and range), named ranges, zoom, density toggle, color coding, character and range jump, pin side by side, asset tagging. Same component on iPad and web.	Can create multiple timelines, add events, navigate and filter them, and tag characters and locations to events on both platforms.
13	Notes: folder tree, rich text editor, quick-capture, full-text search, private notes. Same on iPad and web.	Can create folders, write notes, quick-capture from anywhere, and search across all notes on both platforms.
14	Collaboration: invite by email, Editor and Viewer roles, page status tracking, update banner, 'last edited by' indicator, threaded comments on script blocks.	Two users can collaborate on the same World. Both see page status and whose edit is current. Comments work on both platforms.
15	Script breakdown grid. Script PDF export (server-side Puppeteer, background job, download notification). Read-only share link.	Can see all pages at a glance with status indicators. Can export a script as industry-standard PDF. Can share a read-only link.
16	Polish, bug fixes, Apple Pencil latency tuning, web performance, cross-platform UI consistency audit, onboarding flow for new users.	Core workflow works end-to-end on both platforms with no blocking bugs. A new user can get from sign-up to writing in under 2 minutes.

7. Deferred to v2 — Prioritized

All deferred features are fully defined in the Product Scope document. The order below reflects priority after MVP validation based on likely user feedback.

First wave — immediately after MVP validation

- Storyboard auto-generation from script — the parser-to-canvas pipeline. The data model is ready. This is the next highest-leverage feature after the MVP is proven.
- Real-time collaborative sync (Yjs CRDT) — replaces last-write-wins. Needed as team usage grows.
- World Bible — Factions, Objects/Artifacts, Lore/Concepts entry types.
- The Chalkboard — freeform notecard canvas for story planning. Keyboard and Apple Pencil card modes.
- Timeline lenses — reader knowledge, character knowledge, story order, real-world reference.

Second wave — after first wave is stable

- Relationship Map — living dynamic diagram of character relationships.
- Pacing View — emotional beat curve across an issue or arc.
- Mood / Tone Board — visual reference library per series or arc.
- Continuity Tracker — automatic detection of knowledge, physical state, and relationship inconsistencies.
- Drawing tools expansion — brush, shapes, lasso select, pressure-sensitive eraser.
- Storyboard PDF export — full layered export with layer compositor.
- Series Bible PDF export.

Third wave — mature product

- The Letterer — hand-lettering font creation from the artist's handwriting.
- Branch proposals — artist git-branch workflow for structural page changes.
- Companion Mode — phone as synchronized script reader while drawing on tablet.
- Offline mode with sync on reconnect.
- Android tablet support.
- Version history and named snapshots.
- Reading mode, Presentation mode, Print layout view.

8. Risks & Mitigations

Risk	Likelihood	Mitigation

Timeline feature becomes the focus of development at the expense of the core writing workflow	High	Sprint plan puts script editor (5-6) and World Bible (7-8) before Timeline (11-12). Do not reorder.
Expo for Web produces a noticeably worse experience than a native Next.js app	Medium	Test web performance in Sprint 1. Fallback: thin Next.js shell that imports Expo components.
Apple Pencil latency on React Native Skia is perceptible during drawing	Medium	Test in Sprint 9. Early enough to pivot to a native Skia module or PencilKit bridge if needed.
Shared rich text editor behaves differently on iPad vs web in subtle ways	Medium	Treat cross-platform parity as a first-class test requirement from Sprint 5.
The data model for timelines becomes too complex as lenses are added in v2	Medium	The AssetTimelineTag polymorphic join table and intent tag system are designed specifically to accommodate lenses without schema changes.
Solo build pace slips and MVP takes too long	High	Timebox to 16 sprints. If Sprint 6 script editor does not feel right, stop and reassess before continuing.

9. Earliest Testable Moment

The earliest useful test is at the end of Sprint 6 — when a writer can open PanelSync on an iPad, write a multi-panel script page using all the comic block types, and have it save correctly. At that point you have enough to put it in front of your collaborator and ask: does this feel better than what we were doing?

You do not need the Timeline, the World Bible, or the drawing canvas to get that signal. The script editor alone, working well, is enough to validate the core premise.

Sprint 6 Gate	*If the script editor does not feel as good as a purpose-built comic writing tool by Sprint 6, stop and redesign before proceeding. Everything else in this document builds on that foundation.*

This document should be treated as a living reference. Update it when scope decisions change — do not let the code diverge silently from the plan.